

Milestone 2

Proof of Achievement Report

Project Catalyst #1200045: MPC as a Layer 2 Service to Cardano

Profila AG x Partisia Blockchain

Date: 2026-03-01

Network: Partisia Blockchain Testnet + Cardano Preprod

1. Executive Summary

Milestone 2 delivers the Partisia Blockchain MPC smart contract, the ZK computation kernel, and the off-chain relay that bridges Cardano Layer 1 to Partisia Layer 2. The contract is deployed to Partisia Blockchain testnet and the relay successfully detects Cardano MpcRequest UTxOs and bridges them to the PBC contract.

All 6 acceptance criteria are met.

2. Acceptance Criteria - Evidence Matrix

#	Criterion	Status	Evidence
1	PBC unit tests pass	PASS	Section 4 - 11/11 tests pass
2	Contract executes MPC	PASS	Section 3 - ZK contract + compute kernel
3	PDF documentation	PASS	This document
4	Test results in docs	PASS	Section 4 + test-results-m2.txt
5	YouTube video of demo	PASS	youtu.be/IC_2EG6gxX0
6	Testnet execution on-chain	PASS	Section 5 - PBC Testnet Browser

3. Smart Contract - Technical Summary

3.1 Contract Details

Property	Value
Language	Rust (PBC SDK v.16.126.0)
ZK Compiler	v6.35.0
Contract Address	039ed9214602f2a93eb05411f852fa78a476607634
Source File	partisia/contracts/profila_mpc/src/contract.rs
ZK Compute	partisia/contracts/profila_mpc/src/zk_compute.rs
Test File	partisia/tests/test-results-m2.txt

3.2 Contract State - ProfilaMpcState

The contract maintains public state with the following fields:

Field	Type	Purpose
administrator	Address	Who deployed/administrates the contract

dataset_id	Vec<u8>	Identifies the Profila dataset being queried
query_type	String	"age_threshold" or "survey_match"
min_participants	u32	Minimum secrets required before computation
result	Option<i64>	Aggregate result after MPC completes
computation_complete	bool	Whether computation has finished

3.3 Entry Points

Function	Shortname	Description
initialize()	init	Deploy contract with dataset parameters
submit_secret_value()	0x40	User submits secret (age or survey answer)
on_variable_inputted()	0x41	Callback when secret confirmed on-chain
start_computation()	0x01	Admin triggers MPC computation
on_compute_complete()	0x42	Callback when MPC finishes
on_variables_opened()	-	Reads declassified aggregate result

3.4 ZK Computation Kernel

Two computation functions run securely across 4 MPC nodes - no single node ever sees an individual user's value:

- compute_aggregate (0x61): Counts secret values > 18 (age_threshold query)
- compute_survey_sum (0x62): Sums secret values directly (survey_match query)

4. Unit Tests - Results

Command: cd partisia/contracts && cargo test

Result: 11/11 PASSED

#	Test Name	Result	Category
1	test_init_contract_valid_query_types	ok	Initialisation validation
2	test_init_contract_rejects_invalid_query_types	ok	Initialisation validation
3	test_init_contract_min_participants_validation	ok	Initialisation validation
4	test_compute_age_threshold_result	ok	MPC result decoding
5	test_compute_survey_match_result	ok	MPC result decoding
6	test_decode_zero_result	ok	Edge case
7	test_decode_large_result	ok	Edge case
8	test_decode_short_byte_slice	ok	Edge case
9	test_decode_empty_slice	ok	Edge case
10	test_survey_sum_result_decoding	ok	Survey sum decoding
11	test_query_type_dispatch_logic	ok	Dispatch branching

5. On-Chain Deployment - Evidence

5.1 PBC Testnet Deployment

Property	Value
Contract Address	039ed9214602f2a93eb05411f852fa78a476607634
Deployment TX	d1394ee403db29de119d7b6b9ed9214602f2a93eb05411f852fa78a476607634
Network	Partisia Blockchain Testnet
Deployed At	2026-03-06T16:10:32Z
Gas Balance	332,837
Storage Size	82,864 bytes

5.2 Verification Links

PBC Testnet Browser:

<https://browser.testnet.partisiablockchain.com/contracts/039ed9214602f2a93eb05411f852fa78a476607634>

5.3 Cardano Initiation Evidence

The relay service detected the Cardano MpcRequest UTxO and bridged it to the PBC contract. This demonstrates the cross-chain initiation from Cardano to Partisia:

```
{
  "cardano_tx": "50d28fd2bd263a84...418d53a57",
  "dataset_id": "profil_a_test_v1",
  "query_type": "age_threshold",
  "pbc_contract_address": "039ed9214602f2a9...76607634",
  "relayed_at": "2026-03-01T22:53:16.322Z"
}
```

Note: This is the first known integration between Cardano and Partisia Blockchain. No existing bridges/integrations exist between these chains for MPC, as confirmed by the Partisia tech team. This has been accomplished with Ethereum to an extent, hence our concept project proposal.

6. Cross-Chain Architecture

Cardano Preprod	Off-Chain Relay	PBC Testnet
+-----+	+-----+	+-----+
Aiken SC	Bfrost relay.ts	REST ZK Contract
MpcRequest	-----> poll UTxOs	-----> profila_mpc
inline datum	API parse datum	API .zkwa + .abi
TX: 50d28f..	relay-log	039ed921..
+-----+	+-----+	+-----+
ON-CHAIN	TESTED	DEPLOYED

7. Repository

GitHub: https://github.com/Profila/Aiken-Partisia_MPC

Contract source: `partisia/contracts/profila_mpc/src/contract.rs`

ZK compute: `partisia/contracts/profila_mpc/src/zk_compute.rs`

Test results: `partisia/tests/test-results-m2.txt`

Relay service: `relay/relay.ts`

Relay log: `relay/relay-log.json`

Deployment evidence: `partisia/deploy/deploy-m3.json`

8. SDK & Toolchain Versions

Tool	Version
cargo-partisia-contract	5.513.0
PBC SDK	v.16.126.0 (binder 11.9.0)
ZK Compiler	6.35.0 (Maven JAR)
Rust	1.86.0
Java	OpenJDK 17.0.18
wasm32-unknown-unknown	installed via rustup

9. Video Demo

M2 Demo: https://youtu.be/IC_2EG6gxX0

Combined M1-M4 Demo: <https://youtu.be/jwwfdTSUS2w>

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